

CHAPTER 1

PHYSICAL QUANTITIES, UNITS & MEASUREMENTS

1.1 Physical quantities and measurement techniques

- 1 Describe how to measure a variety of lengths with appropriate precision using tapes, rulers and micrometers (including reading the scale on an analogue micrometer)
- 2 Describe how to use a measuring cylinder to measure the volume of a liquid and to determine the volume of a solid by displacement
- 3 Describe how to measure a variety of time intervals using clocks and digital timers
- 4 Determine an average value for a small distance and for a short interval of time by measuring multiples (including the period of oscillation of a pendulum)
- 5 Understand that a scalar quantity has magnitude (size) only and that a vector quantity has magnitude and direction
- 6 Know that the following quantities are scalars: distance, speed, time, mass, energy and temperature
- 7 Know that the following quantities are vectors: displacement, force, weight, velocity, acceleration, momentum, electric field strength and gravitational field strength
- 8 Determine, by calculation or graphically, the resultant of two vectors at right angles

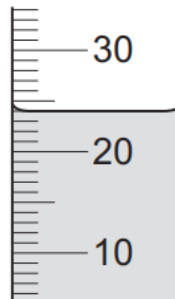
1. O/N/23/11/Q1

Which description is of a scalar quantity?

- A** It has a direction only.
- B** It has direction and unit only.
- C** It has direction, magnitude and unit.
- D** It has magnitude and unit only.

2. O/N/23/12/Q1

The diagram shows the level of liquid in a measuring cylinder. The scale is marked in cm^3 .



What is the volume of the liquid?

- A** 20.4 cm^3 **B** 24.0 cm^3 **C** 24.5 cm^3 **D** 25.0 cm^3

3. M/J/23/11/Q1

A student wishes to measure directly the circumference of a football.

What is the most suitable instrument to use?

- A** a clock only
- B** a measuring tape only
- C** a micrometer only
- D** a ruler only

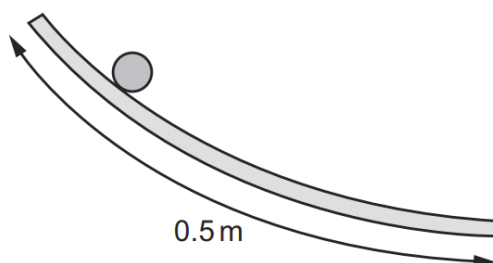
4. M/J/23/11/Q2

Which equation contains two vector quantities?

- A** $\text{acceleration} = \frac{\text{change in velocity}}{\text{time taken}}$
- B** $\text{average speed} = \frac{\text{distance travelled}}{\text{time taken}}$
- C** $\text{density} = \frac{\text{mass}}{\text{volume}}$
- D** $\text{volume} = \text{length} \times \text{width} \times \text{height}$

5. M/J/23/12/Q1

In an experiment, a ball is rolled down a curved track that is about half a metre long.



Which measuring device is used to measure the length accurately?

- A** metre rule
- B** micrometer
- C** stop-watch
- D** tape measure

6. M/J/23/12/Q2

Which row describes acceleration, displacement, distance and speed?

	acceleration	displacement	distance	speed
A	scalar	scalar	vector	scalar
B	scalar	vector	scalar	vector
C	vector	scalar	vector	vector
D	vector	vector	scalar	scalar

7. O/N/22/11/Q2

Velocity is given by the change in displacement divided by the change in time.

How many vector quantities appear in this statement?

- A** 0
- B** 1
- C** 2
- D** 3

8. O/N/22/12/Q3

Which value is one-thousandth of a metre?

- A** 0.0001 cm
- B** 0.001 cm
- C** 0.01 cm
- D** 0.1 cm

9. O/N/21/11/Q1

Which quantity is a vector?

- A** speed
- B** force
- C** mass
- D** distance

10. O/N/21/12/Q2

A student investigates the motion of a ball falling through the air.

Which quantity is a vector?

- A** the diameter of the ball
- B** the gravitational force on the ball
- C** the distance from which the ball is dropped
- D** the speed at which the ball hits the ground

11. M/J/21/11/Q1

A list of various quantities is shown.

acceleration

displacement

force

length

mass

velocity

How many of these quantities are vectors?

- A** 2 **B** 3 **C** 4 **D** 5

12. M/J/20/11/Q2

What is measured using a micrometer?

- A** area
- B** current
- C** length
- D** mass

13. O/N/19/11/Q1

Which word is the name of a scalar quantity?

- A** acceleration
- B** force
- C** mass
- D** velocity

14. O/N/19/11/Q4

What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg / m ³
D	weight	kg

15. O/N/19/12/Q1

Which word is the name of a vector quantity?

- A** density
- B** displacement
- C** energy
- D** speed

16. M/J/19/11/Q1

Which quantities are both vectors?

- A** acceleration and length
- B** distance and velocity
- C** length and distance
- D** velocity and acceleration

17. M/J/19/11/Q4

The magnitudes of three different electric charges are given below.

What is the correct order of size, from largest to smallest?

- A** 1 mC → 1 MC → 1 kC
- B** 1 MC → 1 mC → 1 kC
- C** 1 MC → 1 kC → 1 mC
- D** 1 kC → 1 mC → 1 MC

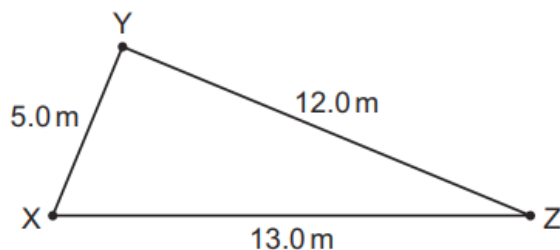
18. M/J/19/12/Q1

Which reading is given to one tenth of a millimetre?

- A** 3.3 cm **B** 3.31 cm **C** 3.310 cm **D** 3.312 cm

19. O/N/18/12/Q9

Paths are laid as shown between points X, Y and Z.



A person walks along the paths from X to Y to Z and then back to X.

What is the value of the total displacement and of the total distance travelled?

	total displacement /m	total distance travelled /m
A	0	0
B	0	30
C	30	0
D	30	30

20. O/N/18/11/Q1

What is a possible mass for a normal adult person?

- A** 7.5 kg **B** 75 kg **C** 750 kg **D** 7500 kg

21. O/N/17/11/Q1

Which list contains only scalar quantities?

- A** acceleration, displacement, velocity
B distance, force, speed
C force, length, time
D length, mass, speed

22. M/J/16/11/Q2

Which set of quantities are all vectors?

- A** acceleration, displacement, velocity
- B** chemical energy, mass, power
- C** extension, force, gravitational potential energy
- D** weight, kinetic energy, work

23. M/J/15/12/Q1

Which quantity is a scalar?

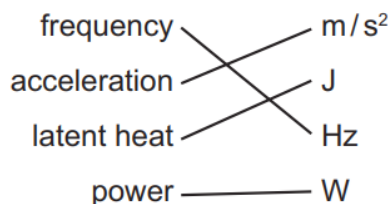
- A** acceleration
- B** force
- C** temperature
- D** velocity

24. O/N/14/11/Q3

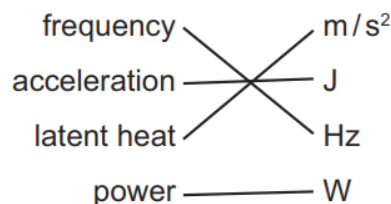
In a test, four students linked the quantities on the left with their units on the right.

Which student matched them all correctly?

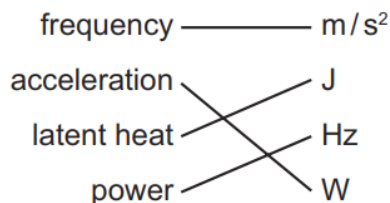
A



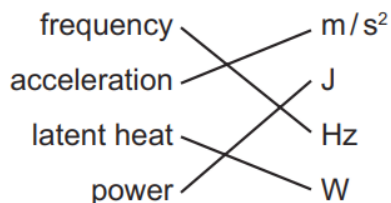
B



C



D



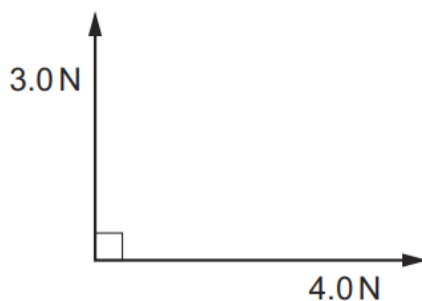
25. O/N/14/11/Q2

Which device can be used to measure the thickness of a single sheet of paper?

- A** a metre rule
- B** a micrometer
- C** a plastic ruler
- D** a measuring tape

26. M/J/14/12/Q2

What is the size of the resultant of the two forces shown in the diagram?



- A** 1.0 N **B** 3.5 N **C** 5.0 N **D** 7.0 N

27. M/J/14/12/Q1

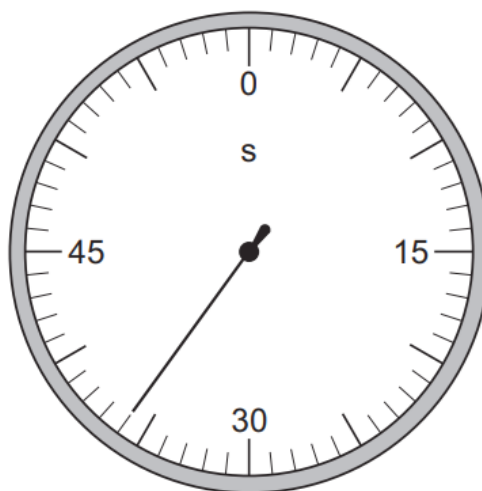
Each row contains a vector and a scalar.

In which row is the size of the vector equal to the size of the scalar?

	vector	scalar
A	displacement of a car	speed of the car
B	velocity of a car	distance travelled by the car
C	velocity of a car	speed of the car
D	weight of a car	mass of the car

28. M/J/14/11/Q2

The diagram shows a stopwatch.



What is the reading on the stopwatch?

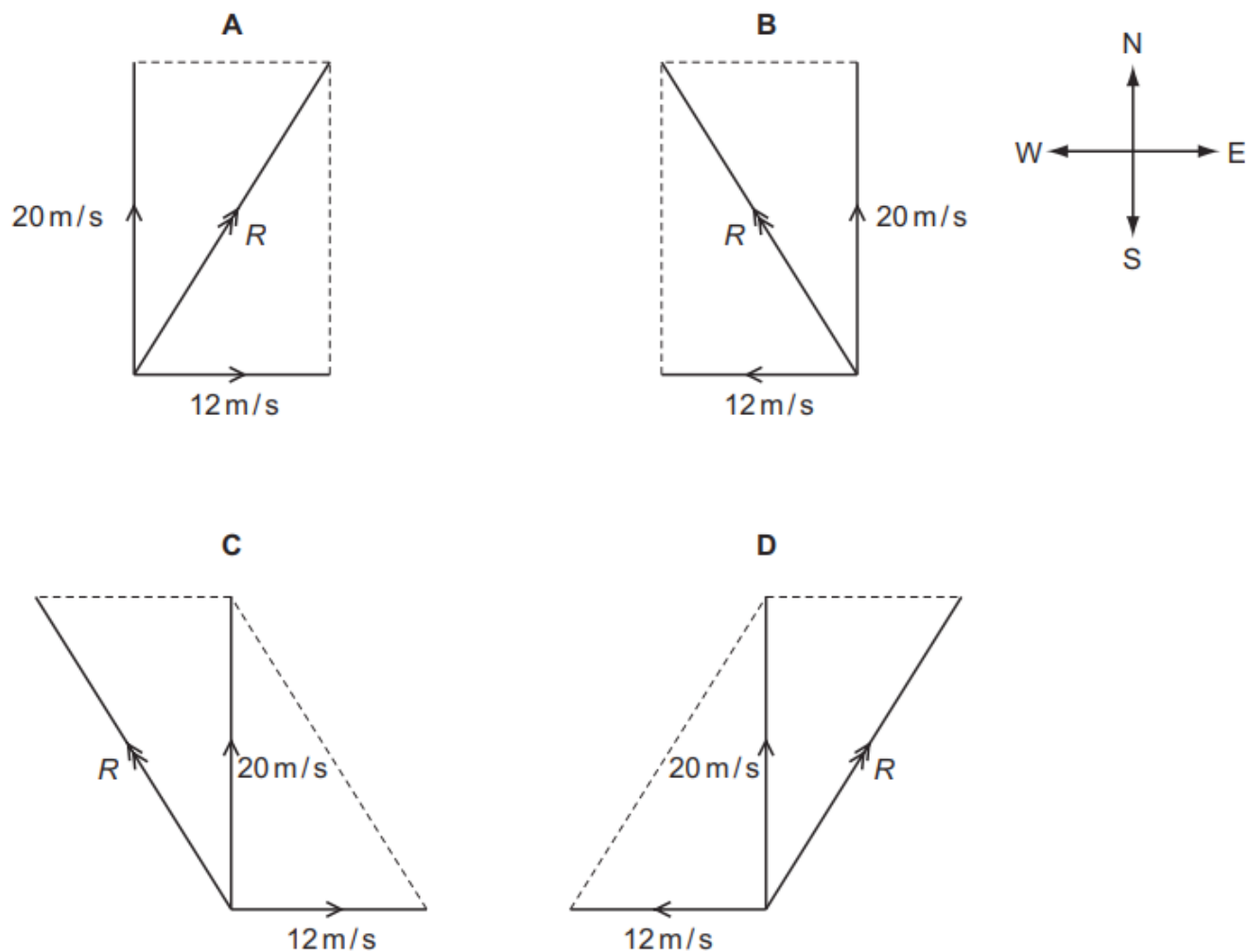
- A** 30.6 s **B** 33.0 s **C** 36.0 s **D** 36.6 s

29. O/N/13/11/Q1

When there is no wind, the engines of an airship push it due north at 20 m/s.

The wind is blowing from the west at 12 m/s.

Which vector diagram correctly shows how the resultant velocity R of the airship is obtained?



30. M/J/13/12/Q2

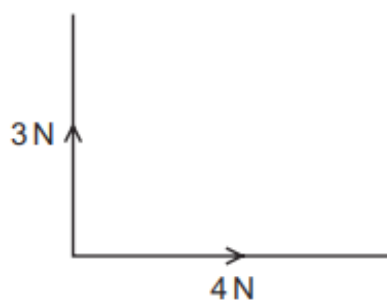
Before marking the finishing line on a running track, a groundsman measures out its 100 m length.

Which instrument is the most appropriate for this purpose?

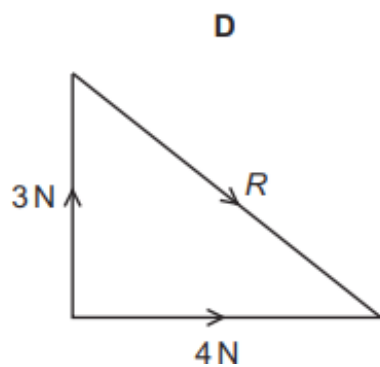
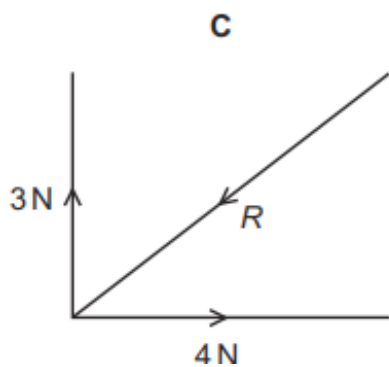
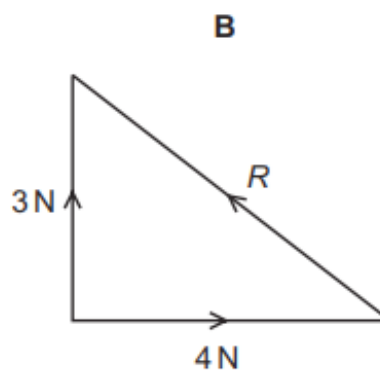
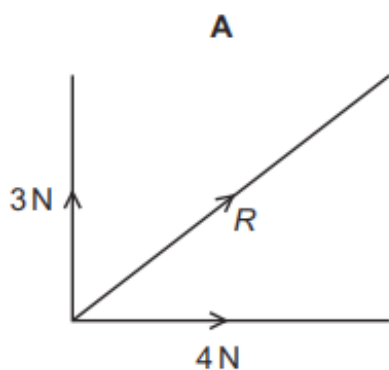
- | | |
|-------------------------|----------------------|
| A measuring tape | C 30 cm ruler |
| B metre rule | D micrometer |

31. M/J/13/11/Q1

Forces of 3 N and 4 N act as shown in the diagram.

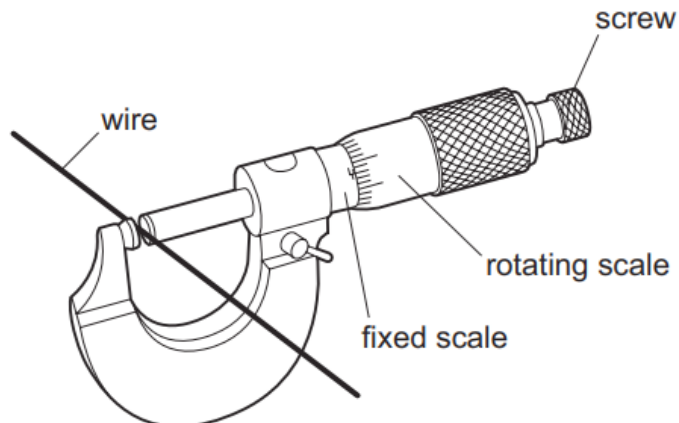


Which diagram shows the resultant R of these two forces?



32. M/J/13/11/Q2

A micrometer is used to measure the diameter of a uniform wire.



What is done to obtain an accurate answer?

- A** Find the reading and add or subtract the zero error.
- B** Make the micrometer horizontal.
- C** Subtract the fixed scale reading from the rotating scale reading.
- D** Subtract the rotating scale reading from the fixed scale reading.

33. M/J/12/12/Q2

Which row correctly shows examples of a vector quantity and a scalar quantity?

	vector	scalar
A	area	force
B	mass	density
C	velocity	acceleration
D	weight	volume

34. O/N/11/12/Q7

A quantity is quoted as having a value of 6.2 ms.

In what units is it measured?

- | | |
|----------------------------|-----------------------|
| A metres | C microseconds |
| B metres per second | D milliseconds |

35. M/J/11/12/Q1

What is the correct unit for the quantity shown?

	quantity	unit
A	electromotive force (e.m.f.)	N
B	latent heat	J
C	pressure	kg / m ³
D	weight	kg

36. M/J/10/11/Q2

The following statements are about motion.

- 1 A plane flies due East for 600 km.
- 2 A runner's average speed in a race around a track is 5 m/s.
- 3 A snail crawls at 3 mm/s in a straight line towards a lettuce.
- 4 A tourist travels 500 km on a journey.

Which statements describe vector quantities?

- A** 1 and 2 **B** 1 and 3 **C** 2 and 3 **D** 2 and 4

37. M/J/10/11/Q1

Power is measured in watts.

What is the correct symbol for millions of watts?

- A** mw **B** mW **C** Mw **D** MW

38. M/J/09/Q2

Which statement about scalars and vectors is correct?

- A** A scalar has direction but no size. **C** A vector has direction but no size.
B A scalar has size but no direction. **D** A vector has size but no direction.

39. O/N/06/Q2

A student studies some equations.

$$\text{power} = \text{work} / \text{time}$$

$$\text{force} = \text{mass} \times \text{acceleration}$$

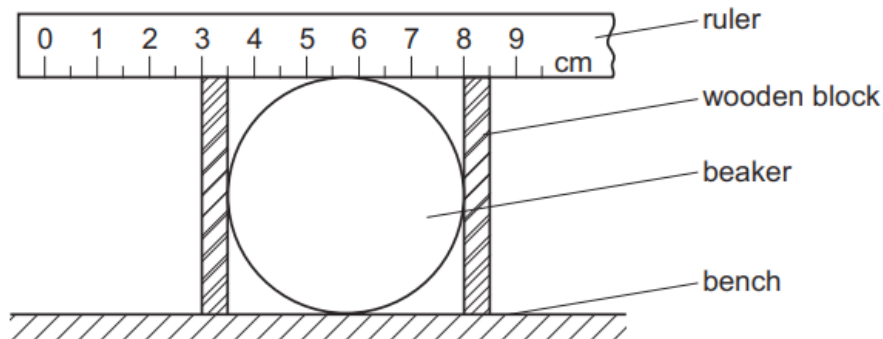
$$\text{velocity} = \text{displacement} / \text{time}$$

How many vector quantities are contained in the equations?

- A** 1 **B** 2 **C** 3 **D** 4

40. O/N/06/Q1

The diagram shows one method of measuring the diameter of a beaker.

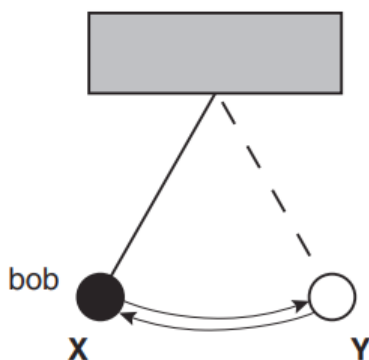


What is the diameter of the beaker?

- A** 4.5 cm **B** 5.0 cm **C** 5.5 cm **D** 8.0 cm

41. O/N/03/Q1

One oscillation of a swinging pendulum occurs when the bob moves from **X** to **Y** and back to **X** again.



Using a stopwatch, which would be the most accurate way to measure the time for one oscillation of the pendulum?

- A** Time 20 oscillations and multiply by 20.
B Time 20 oscillations and divide by 20.
C Time one oscillation.
D Time the motion from **X** to **Y**, and double it.

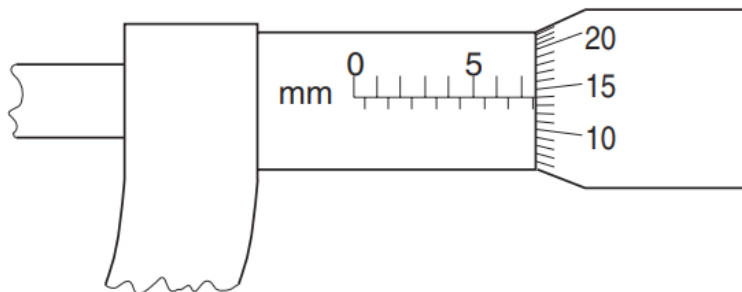
42. M/J/03/Q8

Which of the following groups of physical quantities consists only of scalars?

- | | |
|--|--------------------------------|
| A acceleration, force, velocity | C force, time, velocity |
| B acceleration, mass, speed | D mass, speed, time |

43. O/N/02/Q1

The diagram shows a micrometer scale.

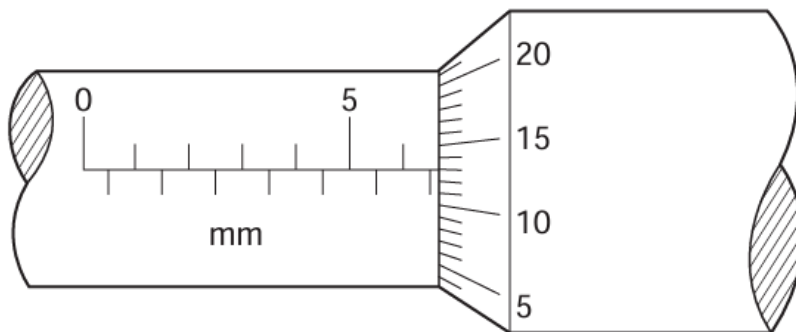


Which reading is shown?

- A** 5.64 mm **B** 7.14 mm **C** 7.16 mm **D** 7.64 mm

44. M/J/08/Q1

What is the reading on this micrometer?



- A** 5.43 mm **B** 6.63 mm **C** 7.30 mm **D** 8.13 mm